WTW 158 Worksheet 2 on Functions and Trigonometry

A, TRIGONOMETRY WORKSHEET on CLICKUP
Do before the prac

B. TEXTBOOK PROBLEMS - do before the prac
p 19 no 25, 34, 35, 43, 76
p A32: 20, 34, 68, 69, 74

C. MORE PROBLEMS

1.1Sketch an **odd** function f such that $f(x) = \begin{cases} 2 & \text{if } 0 < x \leq 1 \\ 2/x & \text{if } 1 < x \leq 3 \end{cases}$

1.2 Find the domain and range of f.
2. Find $(i)\sin 50\pi$ $(ii)\cot\frac{15\pi}{4}$ $(iii)\sec(-\frac{11\pi}{6})$ $(iv)\operatorname{cosec}\frac{11\pi}{2}$
3. Solve for x:
(i) $\sin^2 x = 1, x \in [0, 2\pi]$
(ii)$2\sin^2 x = 3\cos x, x \in [-\pi, \pi]$

4. Solve by means of a graph: $\sin x > \cos x, x \in [0, 2\pi]$
D. TEST QUESTIONS
Question 1
Write f out as a piece-wise defined function and determine the domain and range of the function $f(x) = |1 - 4x^2|$.

Question 2
$\sin(-\frac{13\pi}{2}) =$

[a] 1	[b] -1	[c] 0	[d] None of these

Question 3
If $\sin x = -\frac{1}{3}, \pi < x < \frac{3\pi}{2}$, then $\cot x =$

[a] $\sqrt{8}$	[b] $-\sqrt{8}$	[c] $\frac{1}{\sqrt{8}}$	[d] $\frac{-1}{\sqrt{8}}$	[e] $\frac{3}{\sqrt{8}}$	[f] $\frac{-3}{\sqrt{8}}$	[g] None of these

Question 4
Solve for x if $2\cos x \sin x = -\cos x$.

Question 5
Determine $x \in [0, 2\pi]$ such that $|\cos x| \leq \frac{1}{2}$ by using a sketch.

E YOUTUBE VIDEOS
Search for: PatrickJMT
A way to remember the entire unit circle for trigonometry
Solving trogonometric equations

SELECTED ANSWERS
A. Solutions on ClickUP
B. p19 34. $t \in [-2, 3]$ 35. $x \in \mathbb{R}\backslash[0, 5]$ 76. f is odd. p32 68. $x \in \{\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}\}$,
69. $x \in \{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}, \frac{3\pi}{2}\}$ 74. $x \in [0, \frac{2\pi}{3}) \cup (\frac{4\pi}{3} 2\pi]$
C. 1.2 Domain: $[-3, 3]\backslash\{0\}$ Range: $[-2, -\frac{2}{3}] \cup [\frac{2}{3}, 2]$ 2 (i) 0 (ii) -1, (iii) $\frac{2}{\sqrt{3}}$ (iv) -1 3 .(i) $x \in \{\frac{\pi}{2}, \frac{3\pi}{2}\}$ (ii) $x = \{-\frac{\pi}{3}, \frac{\pi}{3}\}$ 4. $x \in (\frac{\pi}{4}, \frac{5\pi}{4}\}$
D. Q1: Dom $\mathbb{R}$, Range $[0, \infty)$ Q2: [b], Q3: [a] Q4: $x = \frac{\pi}{6} + k\pi, x = \frac{5\pi}{6} + 2k\pi, k \in \mathbb{Z}$,
Q5. $x \in [\frac{\pi}{3}, \frac{2\pi}{3}] \cup [\frac{4\pi}{3}, \frac{5\pi}{3}]$

WTW 158 Worksheet 2.

A. Solutions on ClickUP

B. p 19.

25. $f(x) = 3x^2 - x + 2$

$\Rightarrow f(2) = 12$; $f(-2) = 16$;

$f(a) = 3a^2 - a + 2$

$f(-a) = 3a^2 + a + 2$

$f(a+1) = 3(a+1)^2 - (a+1) + 2$

$2f(a) = 6a^2 - 2a + 2$

$f(2a) = 12a^2 - 2a + 2$

$f(a^2) = 3a^4 - a^2 + 2$

$(f(a))^2 = (3a^2 - a + 2)^2$

$f(a+h) = 3(a+h)^2 - (a+h) + 2$

34. $g(t) = \sqrt{3-t} - \sqrt{2+t}$

Domain: $3 - t \geq 0$ and $2 + t \geq 0$

$\Rightarrow t \leq 3$ and $t \geq -2$

$\Rightarrow t \in [-2, 3]$

35. $f(x) = \dfrac{1}{\sqrt[4]{x^2 - 5x}}$

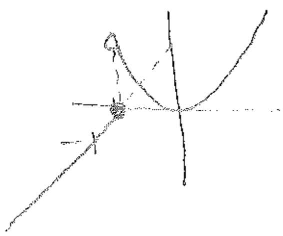
Domain: $x^2 - 5x > 0$

$\Rightarrow x(x-5) > 0$

$\Rightarrow x \in \mathbb{R} \setminus [0, 5]$

43. $f(x) = \begin{cases} x+1, & x \leq -1 \\ x^2, & x > -1 \end{cases}$

$f(-3) = -2$; $f(0) = 0$; $f(2) = 4$



76. $f(x) = x|x|$

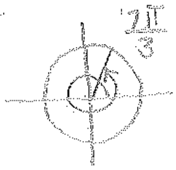
$f(-x) = -x|-x|$

$= -x|x| = -f(x)$

$\Rightarrow f$ is odd

p A32

20.



34. $\operatorname{cosec}\theta = -\frac{4}{3}$

$\Rightarrow \sin\theta = -3/4$

$\cos\theta = \frac{\sqrt{7}}{4}$ $\quad \sec\theta = \frac{4}{\sqrt{7}}$

$\tan\theta = -\frac{3}{\sqrt{7}}$ $\quad \cot\theta = -\frac{\sqrt{7}}{3}$

68. $|\tan x| = 1 \Rightarrow \tan x = \pm 1$

$\Rightarrow x \in \left\{\pi/4, 3\pi/4, \frac{5\pi}{4}, \frac{7\pi}{4}\right\}$

69. $\sin 2x = \cos x$

$\Rightarrow 2\sin x \cdot \cos x = \cos x$

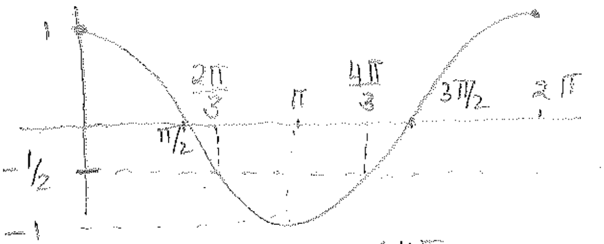
$\Rightarrow (2\sin x - 1)\cos x = 0$

$\Rightarrow \sin x = \frac{1}{2}$ or $\cos x = 0$

$\Rightarrow x = \pi/6, 5\pi/6, \pi/2, \frac{3\pi}{2}$

74. $2\cos x + 1 > 0$

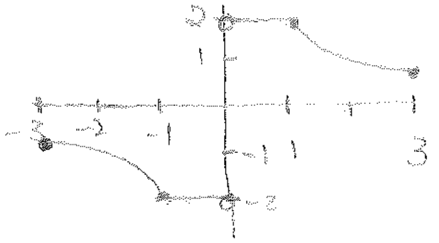
$\Rightarrow \cos x > -1/2$.



$\Rightarrow x \in \left[0, \frac{2\pi}{3}\right) \cup \left(\frac{4\pi}{3}, 2\pi\right]$

C. MORE PROBLEMS

1.1



1.2 Domain : $[-3, 3] \setminus \{0\}$

Range : $[-2, -\frac{2}{3}] \cup [\frac{2}{3}, 2]$

2. (i) $\sin 50\pi = 0$ (ii) $\cot \frac{15\pi}{4} = -1$

(iii) $\sec(-\frac{11\pi}{6}) = \frac{2}{\sqrt{3}}$ (iv) $\operatorname{cosec} \frac{11\pi}{2} = -1$

3. (i) $\sin^2 x = 1$

$\Rightarrow \sin x = \pm 1$

$\Rightarrow x \in \{\pi/2, 3\pi/2\}$

(ii) $2\sin^2 x = 3\cos x$

$\Rightarrow 2(1-\cos^2 x) = 3\cos x$

$\Rightarrow -2\cos^2 x - 3\cos x + 2 = 0$

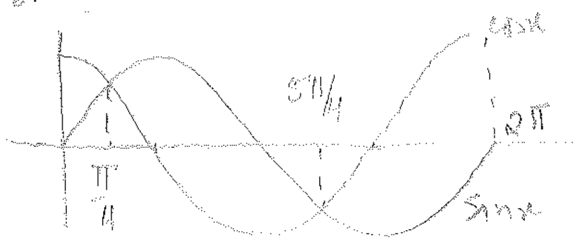
$\Rightarrow 2\cos^2 x + 3\cos x - 2 = 0$

$\Rightarrow (2\cos x - 1)(\cos x + 2) = 0$

$\Rightarrow \cos x = \frac{1}{2}$ or $\cos x = -2$ no sol^n.

$\Rightarrow x = \{-\pi/3, \pi/3\}$

4. $\sin x > \cos x$



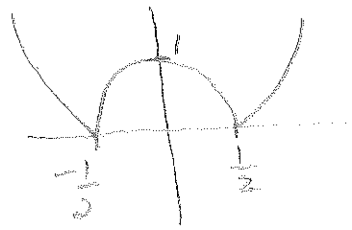
$\Rightarrow x \in (\pi/4, 5\pi/4)$

D. TEST QUESTIONS

Q1

$$f(x) = \begin{cases} 1-4x^2 & \text{if } 1-4x^2 \geq 0 \\ -(1-4x^2) & \text{if } 1-4x^2 < 0 \end{cases}$$

$$= \begin{cases} 1-4x^2 & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ -1+4x^2 & \text{if } x \geq \frac{1}{2} \text{ or } x \leq -\frac{1}{2} \end{cases}$$



Domain: $\mathbb{R}$

Range : $[0, \infty)$.

Q2 [b]

Q3 [a]

$\cot x > 0$

Q4 $2\cos x \sin x + \cos x = 0$

$\Rightarrow \cos x(2\sin x + 1) = 0$

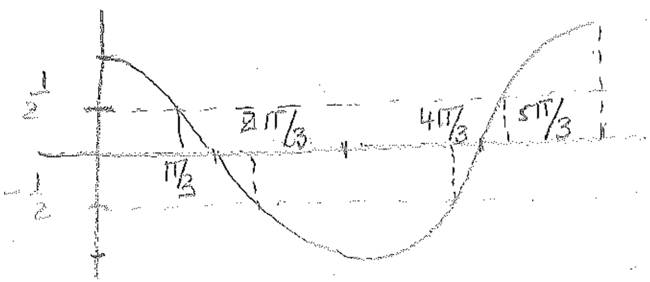
$\Rightarrow \cos x = 0$ or $\sin x = -1/2$

$\Rightarrow x = \frac{\pi}{2} + k\pi$ or $x = \pi/6 + 2k\pi$

or $x = \frac{5\pi}{6} + 2k\pi$

$k \in \mathbb{Z}$

Q5 $|\cos x| \leq \frac{1}{2} \Rightarrow -\frac{1}{2} \leq \cos x \leq \frac{1}{2}$



$x \in [\pi/3, 2\pi/3] \cup [\frac{4\pi}{3}, \frac{5\pi}{3}]$